Theorem 1. *Consider the multi-AUV system (7) subjected to bounded disturbances $\boldsymbol{d}_i$. Suppose the FTESO is formulated as (9). If the gain parameters $l_{i,j}$ and $k_{i,j}$ are chosen such that the characteristic polynomials $s^3+l_{i,1}s^2+l_{i,2}s+l_{i,3}$ and $s^3+k_{i,1}s^2+k_{i,2}s+k_{i,3}$ are strictly Hurwitz, then the estimation errors $\boldsymbol{e}_{i,1},\boldsymbol{e}_{i,2},\boldsymbol{e}_{i,3}$ will converge to a bounded compact region containing the origin in practical finite time.*

Proof. To facilitate the rigorous derivation, the error dynamics (10) are decoupled into a nominal fractional homogeneous part $\boldsymbol{f}_{hom}(\boldsymbol{e}_i)$, a linear correction part $\boldsymbol{f}_{lin}(\boldsymbol{e}_i)$, and a bounded perturbation part $\boldsymbol{\Delta}_i$, such that

$$\dot{\boldsymbol{e}}_i = \boldsymbol{f}_{hom}(\boldsymbol{e}_i) + \boldsymbol{f}_{lin}(\boldsymbol{e}_i) + \boldsymbol{\Delta}_i \tag{11}$$

where $\boldsymbol{e}_i = [\boldsymbol{e}_{i,1}^T, \boldsymbol{e}_{i,2}^T, \boldsymbol{e}_{i,3}^T]^T \in \mathbb{R}^{15}$, $\boldsymbol{\Delta}_i = [\boldsymbol{0}_{1\times5}, \boldsymbol{0}_{1\times5}, -\boldsymbol{\sigma}_i^T]^T$, and

$$\boldsymbol{f}_{hom}(\boldsymbol{e}_i) = \begin{bmatrix} \boldsymbol{e}_{i,2} - l_{i,1}\lfloor \boldsymbol{e}_{i,1}\rceil^{\alpha_1} \\ \boldsymbol{e}_{i,3} - l_{i,2}\lfloor \boldsymbol{e}_{i,1}\rceil^{\alpha_2} \\ -l_{i,3}\lfloor \boldsymbol{e}_{i,1}\rceil^{\alpha_3} \end{bmatrix}, \quad \boldsymbol{f}_{lin}(\boldsymbol{e}_i) = \begin{bmatrix} -k_{i,1}\boldsymbol{e}_{i,1} \\ -k_{i,2}\boldsymbol{e}_{i,1} \\ -k_{i,3}\boldsymbol{e}_{i,1} \end{bmatrix}$$

Primarily, we analyze the unperturbed fractional subsystem $\dot{\boldsymbol{e}}_i = \boldsymbol{f}_{hom}(\boldsymbol{e}_i)$. By defining a dilation weight vector $\mathbf{r} = (1, \alpha, 2\alpha - 1)$, it is straightforward to verify that $\boldsymbol{f}_{hom}(\lambda^{\mathbf{r}}\boldsymbol{e}_i) = \lambda^{\kappa+\mathbf{r}}\boldsymbol{f}_{hom}(\boldsymbol{e}_i)$ holds for any scalar $\lambda > 0$, confirming a homogeneity degree of $\kappa = \alpha - 1 < 0$. Notably, when $\alpha = 1$, the subsystem $\dot{\boldsymbol{e}}_i = \boldsymbol{f}_{hom}(\boldsymbol{e}_i)$ degenerates into a linear system whose characteristic polynomial is exactly $s^3 + l_{i,1}s^2 + l_{i,2}s + l_{i,3}$. Since this polynomial is chosen to be strictly Hurwitz, the linear counterpart is asymptotically stable. According to the finite-time stability theorem for homogeneous systems [13], the continuous extension to a negative degree ($\kappa < 0$) guarantees that $\dot{\boldsymbol{e}}_i = \boldsymbol{f}_{hom}(\boldsymbol{e}_i)$ is globally finite-time stable.

Consequently, there exists a continuously differentiable and homogeneous Lyapunov function $V(\boldsymbol{e}_i)$ of degree $\mu > \max(1, \alpha, 2\alpha - 1)$ such that

$$\beta_1\|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu} \le V(\boldsymbol{e}_i) \le \beta_2\|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu}, \quad \frac{\partial V}{\partial \boldsymbol{e}_i}\boldsymbol{f}_{hom}(\boldsymbol{e}_i) \le -\beta_3\|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu+\kappa} \tag{12}$$

where $\beta_1, \beta_2, \beta_3$ are positive constants, and $\|\boldsymbol{e}_i\|_{\mathbf{r}}$ represents the homogeneous norm.

Subsequently, we consider the full perturbed system. Taking the time derivative of $V(\boldsymbol{e}_i)$ along (11) yields

$$\dot{V} = \frac{\partial V}{\partial \boldsymbol{e}_i}\boldsymbol{f}_{hom}(\boldsymbol{e}_i) + \frac{\partial V}{\partial \boldsymbol{e}_i}\boldsymbol{f}_{lin}(\boldsymbol{e}_i) + \frac{\partial V}{\partial \boldsymbol{e}_{i,3}}(-\boldsymbol{\sigma}_i) \tag{13}$$

Due to $\alpha \in (2/3, 1)$, the linear correction $\boldsymbol{f}_{lin}$ dominates the dynamics for large estimation errors. Since its associated polynomial $s^3 + k_{i,1}s^2 + k_{i,2}s + k_{i,3}$ is strictly Hurwitz, it inherently yields $\frac{\partial V}{\partial \boldsymbol{e}_i}\boldsymbol{f}_{lin} \le 0$, guaranteeing global exponential convergence far from the origin. Conversely, in the small-error region, the fractional term $\boldsymbol{f}_{hom}$ with a negative homogeneity degree ($\alpha - 1 < 0$) mainly

dominates. By conservatively omitting the stabilizing linear term and noting that $\frac{\partial V}{\partial \boldsymbol{e}_{i,3}}$ is homogeneous of degree $\mu - \mathbf{r}_3 = \mu - (2\alpha - 1)$, there exists a constant $\beta_4 > 0$ such that $\left\| \frac{\partial V}{\partial \boldsymbol{e}_{i,3}} \right\| \leq \beta_4 \|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu - 2\alpha + 1}$. Therefore, (13) can be upper bounded by

$$\begin{aligned} \dot{V} &\leq -\beta_3 \|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu+\alpha-1} + \left\| \frac{\partial V}{\partial \boldsymbol{e}_{i,3}} \right\| \|\boldsymbol{\sigma}_i\| \leq -\beta_3 \|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu+\alpha-1} + \beta_4 \bar{\sigma} \|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu-2\alpha+1} \\ &= -\|\boldsymbol{e}_i\|_{\mathbf{r}}^{\mu-2\alpha+1} \left(\beta_3 \|\boldsymbol{e}_i\|_{\mathbf{r}}^{3\alpha-2} - \beta_4 \bar{\sigma} \right) \end{aligned} \tag{14}$$

To guarantee that the stabilizing term inherently dominates the perturbation term near the origin, the power condition must satisfy $(\mu+\alpha-1)-(\mu-2\alpha+1) > 0$, which yields $3\alpha-2 > 0$, i.e., $\alpha > 2/3$. This mathematically justifies the parameter design requirement $\alpha \in (2/3, 1)$.

Therefore, $\dot{V} < 0$ is guaranteed as long as $\|\boldsymbol{e}_i\|_{\mathbf{r}} > (\beta_4 \bar{\sigma} / \beta_3)^{\frac{1}{3\alpha-2}}$. Consequently, driven by the global exponential stability far from the origin and the finite-time robust convergence near the origin, the estimation error $\boldsymbol{e}_i$ achieves practical finite-time stability and will converge into a bounded compact region $\Omega_{\boldsymbol{e}} = \{\boldsymbol{e}_i \mid \|\boldsymbol{e}_i\|_{\mathbf{r}} \leq (\beta_4 \bar{\sigma} / \beta_3)^{\frac{1}{3\alpha-2}}\}$. This completes the proof. ■